MIL-R-12551D

8 April 1986
SUPERSEDING
MIL-R-12551C
4 October 1971

MILITARY SPECIFICATION

RECHARGING UNIT, CARBON DIOXIDE, PUMPING METHOD

This specification is approved for use by all Department and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers an electric-motor-driven, pumping method, carbon dioxide recharging unit.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 3655

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SPECIFICATIONS

FEDERAL

CC-M-1807	- Motors, Alternating Current, Fractional and Integral Horsepower (500 HP and Smaller).
RR-C-901	- Cylinders, Compressed Gas: High Pressure, Steel DOT 3AA, and Aluminum Applications, General Specification for.
TT-P-664	- Primer Coating, Synthetic, Rust-Inhibiting, Lacquer Resisting.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Box, Shipping, Fiberboard.
PPP-P-40	- Packaging and Packing of Hand Tools.

MILITARY

MIL-P-116	- Preservation, Methods of.
MIL-V-173	- Varnish, Moisture-and-Fungus Resistant (For Treatment of Communications, Electronic, and Associated Equipment).
MIL-P-514	- Plates, Indentification, Instruction and Marking, Blank.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-C-2212	- Controllers, Electric Motor A.C. or D.C., and Associated Switching Devices.
MIL-G-3859	- Grease Guns, Hand-Operated, Lever, Push and Screw Type.
MIL-G-10924	- Grease, Automotive and Artillery.
MIL-S-12514	- Starters, Motor; Across-The-Line and Reduced Voltage Types, A-C Induction, 3/4 to 100 HP, Manual and Electric.
MIL-E-16298	- Electric Machines Having Rotating Parts and Associated Repair Parts: Packaging of.
MIL-M-17060	- Motors, 60-Hertz, Alternating Current, Integral-Horsepower, Shipboard Use.
MIL-M-17413	- Motors, Direct Current, Integral H.P., Naval Shipboard.
MIL-M-17556	- Motor, Direct-Current, Fractional HP (Shipboard Use).
MIL-G-20241	- Gasket Material, Wool Felt, Impregnated, Adhesive, Pressure-Sensitive.
MIL-L-21260	- Lubricating Oil, Internal Combustion Engine, Preservative and Break-In.
MIL-G-23827	- Grease, Aircraft and Instrument, Gear and Actuator Screw, NATO Code Number G-354, Metric.
MIL-S-81733	- Sealing and Coating Compound, Corrosion Inhibitive.

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STANDARDS

FEDERAL

FED-STD-H28	- Screw Thread Standards for Federal Services.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-461	- Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1186	- Cushioning, Anchoring, Bracing, Blocking and Waterproofing; with Appropriate Test Methods.
MIL-STD-1472	- Human Engineering Design Criteria for Military Systems, Equipment and Facilities.
MIL-STD-1474	- Noise Limits for Army Materiel.
MS15003	- Fitting, Lubrication (Hydraulic) Surface Check, 1/8 Pipe Threads, Steel, Type III.
MS35844	- Plug, Machine Thread, Magnetic: Drain.

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN IRON AND STEEL INSTITUTE (AISI)

AISI - Steel Products Manual-Stainless and Heat Resisting Steels.

(Applications for copies should be addressed to the American Iron and Steel Institute, 1000 Sixteenth Street, NW, Washington, D.C. 20036.)

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THE AMERICAN SOCIETY OF MECHANICAL ENGINEERS, INC. (ASME)

ANSI/ASME - Boiler and Pressure Vessel Code - Welding and Brazing Qualifications, Section IX.

(Application for copies should be addressed to the American Society of Mechanical Engineers, Inc., 345 East 47th Street, New York, NY 10017.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951 - Standard Practice for Commercial Packaging.

(Application for copies should be addressed to American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN WELDING SOCIETY, INC. (AWS)

AWS B2.1 - Standard for Welding Procedure and Performance Qualification.

(Application for copies should be addressed to the American Welding Society, Inc., 550 Northwest Le Jeune Road, Miami, FL 33126.)

COMPRESSED GAS ASSOCIATION, INC. (CGA)

CGA - Pamphlet S-1.1 Pressure Relief Devices Standards - Part 1 - Cylinders for Compressed Gases.

(Application for copies should be addressed to the Compressed Gas Association, Inc., Crystal Gateway 1, Suite 501, 1235 Jefferson Davis Highway, Arlington, VA 22202.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

MG 1 - Motors and Generators

(Applications for copies should be addressed to the National Electrical Manufacturers Association, 2101 L Street, NW, Suite 300, Washington, DC 20037.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA No. 70 - National Electrical Code.

(Application for copies should be addressed to the National Fire Protection Association, Battery March Park, Quincy, MA 02269.)

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THE RUBBER MANUFACTURERS ASSOCIATION, INC. (RMA)

RMA - Engineering Standards for Multiple V-Belt Drives.

(Application for copies should be addressed to the Rubber Manufacturers Association, Inc., 1400 K Street, NW Washington, DC 20005.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, (except for associated detail specifications, specifications sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The pumping method carbon dioxide recharging unit, hereinafter referred to as "unit", shall consist of a pump, motor and all necessary piping, hose, valves, and screens required to transfer liquid carbon dioxide. The unit shall be base mounted.

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.3 Materials. Materials shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.3.1 Material deterioration and control. The unit shall be fabricated from compatible materials, inherently corrosion and deterioration resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable storage and operating environments to which the unit may be exposed.

3.3.1.1 Dissimilar metals. Dissimilar metals, as defined in MIL-STD-889, shall be electrically insulated from one another to minimize or prevent galvanic corrosion. Insulation may be provided by an insulating barrier such as a corrosion inhibiting sealant conforming to MIL-S-81733 or chromate tape conforming to MIL-G-20241.

3.3.1.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with components and sub-components, and shall make information available, upon request, to the contracting officer or designated representative.

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3.3.2 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the unit may be newly fabricated from recovered materials to the maximum extent practicable, provided the unit produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the unit.

3.3.3 Corrosion-resistant material. Where corrosion-resistant material is specified, corrosion-resistant steel conforming to AISI type 301 through 348 as applicable, copper, brass, bronze, silver, beryllium-copper, copper-nickel alloys, chromium-nickel or aluminum shall be used.

3.4 Design. The unit shall consist of a metal base, piston-type pump, electric motor with speed reduction drive, service manifold, hoses and fittings. Unless otherwise specified (see 6.2), a tilt rack for securing and inverting 50 pound (lb) liquid carbon dioxide storage cylinders and a toolbox shall be supplied with the unit.

3.4.1 Screw threads. Threads shall conform to FED-STD-H28. All fasteners shall be installed with a positive means against working loose. Locking shall be by cotter pins, locknuts, or positive lockwashers. All exposed fasteners shall be of corrosion-resistant material or treated to resist corrosion.

3.4.2 Lubrication. All surfaces requiring lubrication shall be provided with a means for lubricating.

3.5 Reliability and maintainability.

3.5.1 Reliability. The unit shall demonstrate a reliability of 0.90 at the 90 percent confidence level for 8 hours of operation when tested as specified in 4.5.2.3.

3.5.2 Maintainability. The unit shall have a minimum time between scheduled maintenance of 100 hours. Recommended adjustments as established by a maintenance schedule submitted by the contractor prior to test shall not be considered maintenance actions. Time required for scheduled maintenance shall not exceed 4 manhours per hundred hours of operation. All assemblies and installed attachments shall be accessible for servicing, repair and replacement using specified tools. Sizes and types of connectors, plugs, and valves shall be held to a minimum.

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3.5.2.1 Lubrication fittings. Lubrication fittings shall conform to MS15003. Fittings shall be located in a protected position and shall be accessible to a grease gun conforming to MIL-G-3859. Accessibility to fittings shall be provided without the removal or adjustment of accessories or parts. Panels and plates equipped with hand-operable, quick-disconnect fasteners may be removed to provide accessibility.

3.5.2.1.1 Lubricant pressure release. A pressure release device shall be provided where the use of pressure lubricating equipment could damage grease seals or other parts.

3.5.2.2 Lubricant reservoir. Enclosures such as gearcases and transmission housings that contain a reservoir of lubricants for the lubrication of the parts enclosed shall be equipped with dipsticks, check plugs not less than 0.500-inch pipe size, or, sight gages to determine the level of the lubricant. Each enclosure shall be equipped for filling the enclosure with a lubricant and for draining. The drain outlet shall be fitted with a drainplug conforming to MS35844. The drain outlet shall be located so that removal of the plug will result in complete drainage of the lubricant from the enclosure. Drainage shall be into a collecting vessel when the unit is in its normal position. Integral tubes or troughs may be used to convey the lubricant from the drain outlet into a collecting vessel.

3.5.2.3 Instruction plates. Each unit shall be equipped with instruction plates or diagrams, including warnings and cautions, describing any special or important procedures to be followed in assembling, operating, or servicing the unit. Instruction plates shall conform to MIL-P-514, type III, composition C, of type I, grade A, class 1 material. The plates shall be attached by screws, bolts, or rivets in a conspicuous protected location. When specified (see 6.2), commercial instruction plates shall be furnished.

3.6 Performance. The unit shall pump liquid carbon dioxide at a rate of not less than 4 lbs per minute in continuous operations against a pressure of 1000 lbs per square inch gage (psig) above inlet pressure and at a maximum back pressure of 1800 psig in an ambient temperature of not less than 65° F (18.3° C) or more than 75° F (23.9° C). The pump shall be directly driven or shall be connected to the motor through a V-belt or gear reduction train. Provision shall be made for adjusting the belt tension without removing components or parts. The pressurized components of the unit shall withstand a proof pressure of 3,000 psig.

3.7 Human Factors Engineering. The unit shall conform to accepted Human Factors Engineering Design Criteria as described in MIL-STD-1472. Special design emphasis shall be given, but not limited to, paragraphs 4 (General Requirements), 5.1 (Control/Display Integration), 5.2 (Visual Displays), 5.4 (Controls), 5.5 (Labeling), 5.6 (Anthropometry), 5.7 (Workspace Design Requirements), 5.9 (Design for Maintainability), and 5.13 (Hazards and Safety), of MIL-STD-1472, as applicable.

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3.8 Safety. All parts which are energized electrically shall be insulated, fully enclosed, or guarded. All moving parts which are of such nature or so located as to be a hazard to operating or maintenance personnel shall be fully enclosed or guarded. Protective devices shall not impair the operating functions. Safety relief devices shall be vented in a direction so that the discharge will not be directed toward the operating personnel.

3.8.1 Safety disk. A frangible safety disk shall be provided to release the pressure from the cylinder head within the range of 2,650 to 3,000 psig. The disk shall be readily replaceable. A guard shall be provided to deflect flying particles in case of a ruptured disk. The safety disks shall conform to the Compressed Gas Association, Pamphlet S-1.1, type GG-1.

3.8.2 Pump bypass. The pump shall be provided with an external permanently mounted or an internal bypass system containing a normally closed pressure release valve to protect the recharging system against damage, from blockage of charging hose or overcharging cylinders, while the pump is running. The bypass system shall operate with a 1,000 psig plus or minus 100 psig differential above inlet pressure and shall circulate the maximum output of the pump without raising the pressure at any point in the system more than 1,100 psig above the supply pressure.

3.9 Environmental requirements.

3.9.1 Fungus and moisture resistance. The electrical circuitry, including all components and connections except as specified below, shall be protected from the effects of fungus growth and moisture by an overall threatment with a varnish conforming to MIL-V-173, composition as specified (see 6.2) with 1 percent copper 8-quinolinolate (by weight) based on the nonvolatile content of the varnish:

a. Components or circuit elements that are inherently fungus and moisture resistant or which are hermetically sealed need not be treated.

b. Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.

When used, the varnish shall be applied by spray, brush, or a combination of both to give a minimum dry-film thickness of 1 mil to component or element surfaces previously cleaned and prepared so that the surfaces are free from all foreign matter which would interfere with the adherence or function of the varnish.

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3.10 Detailed requirements.

3.10.1 Base mounting. The base mounting shall be fabricated from steel frame members and deck plate or stamped and formed steel plate or cast iron. Holes shall be provided in the base deck for bolting the components of the unit in place. For units having a removable tilt rack attached to the base, provisions shall be made in the base frame for mounting.

3.10.2 Pump. The pump shall be a single or multiple cylinder, single action reciprocating piston-type, designed for continuous delivery of liquid carbon dioxide against a pressure of 1,000 psig above inlet pressure and at a maximum back pressure of 1,800 psig. The pump shall have a rate of not less than 4 lbs of liquid carbon dioxide per minute when operating in an ambient temperature of plus 70° F. (21° C), ±1.5° F (±2.8° C). The pump shall perform, as specified herein, without an interruption in operation for not less than 100 hours.

3.10.2.1 Pump cylinder. The pump cylinder block shall be removable from the crankcase or pump frame for repair or replacement. Packing shall be provided at the point where the piston assembly enters the cylinder bock. An adjustable gland nut or plate shall be provided for adjusting cylinder packing against leaks. A removable cylinder head shall be furnished. The cylinder head shall contain the outlet valve and the pressure-safety-disc-nut. The liquid inlet valve shall be included in the cylinder head or in the piston head.

3.10.2.2 Pump crankcase. When a pump having a closed crankcase is furnished, provisions shall be made to seal against carbon dioxide entering the crankcase.

3.10.3 Hoses and service manifolding. Hoses and service manifolding including all fittings necessary for adapting the storage cylinders and service cylinders through the service hoses to the inlet and outlet of the pump shall be furnished. Swivel adapters having internal threads .830 - 14 NGO - RH - INT, shall be furnished for connection to the service and storage cylinders. The adapter tailpiece at the charging cylinder shall be provided with a strainer of not less than 60 mesh. A 0.250-inch liquid shutoff valve shall be furnished between the pump discharge and the service cylinder. Another 0.250-inch valve shall be furnished as a pressure blowdown to relieve pressure for disconnection of the service cylinder. An optional 0.250-inch valve may be furnished between the pump inlet and the storage cylinder to relieve pressure for disconnection of the storage cylinder. The valve may be in line or manifolded and mounted on the unit. When mounted on the unit, all plumbing and fittings shall be securely attached and rubber grommets shall be provided at each mounting point. Swivel-nut adapters shall be provided at both inlet and outlet ends of each hose, and where required at valves that cannot be rotated in place for removal.

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3.10.3.1 Hoses. Two 6-foot lengths of hose for carbon dioxide service shall be furnished. The hoses shall be corrosion-resistant steel wire reinforced rubber or corrugated nonferrous metal tube. The hoses shall have an external corrosion-resistant metallic wrapping or flexible bronze jacket. The hoses shall have a minimum burst pressure of 5,000 psig.

3.10.3.2 Fittings. All fittings furnished shall be of corrosion-resistant material and shall withstand an internal pressure of at least 3,000 psig. Fitting sizes shall be as applicable. Fittings for adapting service cylinders to be charged to the charging swivel coupling shall be furnished as specified.

3.10.4 Motor and speed reduction assembly. A motor and speed reduction assembly consisting of the motor and associated speed reduction gears, or belts and pulleys shall be furnished to drive the carbon dioxide pump. Direct drive pumps may be furnished if provided with a safety overload switch. When a V-belt drive is furnished, the bilts shall be in accordance with Rubber Manufacturers Association (RMA) engineering standards for belt drives and have a service factor of 1.5.

3.10.4.1 Motor. Unless otherwise specified (see 6.2), the motor shall conform to CC-M-1807, type I, class 1, style A, kind 2, group B, design A or B, continous operation, 1,800 rpm, 115/230 volts (v) ac, 60 Hertz (Hz). All electrical wiring shall be in accordance with NFPA No. 70 for continuous duty operating motors. The continuous duty rating of the motor shall be not less than the driven load when the pump is operated at rated capacity. The motor shall be initially wired for 115-volt service with necessary instructions for converting to 230 volts. When specified (see 6.2), the motor shall be 220-volts or 440-volts ac in accordance with MIL-M-17060 or shall be 115-volts or 230-volts dc in accordance with MIL-M-17556 or MIL-M-17413, service C, ambient temperature rating not less than 40° C., dripproof enclosure, horsepower and speed as required by pump, continuous duty, winding squirrel cage induction for ac motors and stabilized shunt for dc motors, ball bearing, and insulation class A or B.

3.10.4.2 Speed reducer. A directly connected, completely enclosed, gear or a belt-type speed reduction assembly shall be provided to adapt the electric motor shaft speed to the speed required for operating the pump under full load. The motor and pump shall be held stationary by bolts extended through the pump base floor.

3.10.4.3 Power supply cable. Unless otherwise specified (see 6.2), a power supply cable shall be furnished. The cable shall be of suitable size for operation at 115-volt service in accordance with National Electric Code, type SO, two conductor and ground, 25 feet long. The two power conductors shall be color coded one white and one black. The ground wire shall be green. One end of the cable shall be permanently attached to the starting switch on

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the unit. The other end shall be provided with a commercial three-prong grounding type plug for 115-volt service. Provisions shall be made for stowing the power supply cable on the unit frame or for wrapping it on the motor and pump assembly when not in use or during storage and transport. When specified (see 6.2), an appropriate power supply cable for the voltage and power required by the motor specified in 3.10.4.1 shall be furnished.

3.10.4.4 Starting switch. A starting switch with thermal overload protection of sufficient size to service the pump motor shall be furnished. The enclosure containing the starter and overload shall be NEMA type 1 and shall be rigidly mounted on the unit in a location that will permit hand and visual access to all controls. The starter shall conform to MIL-S-12514, type I, class 1, size as applicable. The switch shall be marked to indicate the "ON" and "OFF" positions. When specified (see 6.2), the starter shall conform to MIL-C-2212, ambient temperature rating not less than 40° C., dripproof enclosure, manual operation, across the line, continuous duty, nonautomatic, with thermal overload protection.

3.10.5 Tilt rack. Unless otherwise specified (see 3.4 and 6.2), a tilt rack shall be provided to hold a 50-lb carbon dioxide, storage cylinder conforming to RR-C-901 and invert to a position not more than 30 degrees from the vertical. An adjustable device shall be furnished for rigidy securing storage cylinders of varying diameters in the tilt rack. The tilt rack shall be fabricated from steel and assembled with threaded fasteners or welded. When the tilt rack is a part of the unit base, provisions shall be made for removal during transport or while in storage.

3.10.6 Toolbox. Unless otherwise specified (see 3.4 and 6.2), the unit shall be provided with a weatherproof toolbox of metal not less than 0.071-inch in nominal thickness. The toolbox shall be mounted on the pump base or on the tilt rack frame and shall have a hinged lid and a trunk drawbolt. A drain hole shall be provided. The toolbox shall be of a size to hold the maintenance tools, repair parts, and adapters. The lid shall be self-supporting in the open position.

3.10.7 Lubricants. The unit shall operate as specified herein when lubricated with military lubricants (see 6.4).

3.10.7.1 Grease lubricants. Unless otherwise specified (see 6.4), all grease lubrication, including lubrication of sealed bearings, shall be with greases as follows:

a. MIL-G-10924 - For bearings where operating temperatures do not exceed 175° F.

b. MIL-G-23827 - For instruments, all sealed bearings, and other bearings where operating temperatures do not exceed 250° F.

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The unit shall be assembled, run-in (broken-in), tested, and delivered with these greases. The bearings shall be cleaned before lubrication, because military greases are not always compatible with other greases. Each component shall be tagged in a conspicuous place to indicate which military grease has been used.

3.10.8 Noise limits. The noise produced by the unit shall conform with MIL-STD-1474 requirements when tested in accordance with specification, paragraph 4.5.2.6. The provisions of MIL-STD-1474, paragraphs 4.3 and 4.4 shall be provided if and only if MIL-STD-1474, paragraph 5.1.1.2 procedures have been pursued and documented to the satisfaction of the procurring activity and written permission to exceed the 85 dB(A) limit is obtained from the procuring activity. Hazard signs shall conform with MIL-STD-1474, paragraph 4.3 and be readable at the distance indicated on the sign.

3.11 Electromagnetic interference. Unless otherwise specified (see 6.2 and 6.5), the electromagnetic interference emission of the unit shall conform to MIL-STD-461, class B with intermittent, short duration interference acceptable.

3.12 Color and finish. All parts of the unit and its components normally painted shall be treated and painted in accordance with MIL-T-704, type A, color as specified (see 6.2). Where dissimilar metals are used, all contact surfaces shall be treated and painted as specified in MIL-T-704, type A.

3.13 Identification marking. The unit shall be identified in accordance with MIL-STD-130. Marking shall be applied on plates conforming to MIL-P-514, type I, style 1, composition C of type I, grade A, class 1 material. Plates shall be attached by screws, bolts, or rivets in a conspicuous protected location. When specified (see 6.2), commercial markings shall be used.

3.14 Workmanship.

3.14.1 Metal fabrication. Metals used in the fabrication of the recharging unit shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause injury to the metal. Shearing and chipping shall be done neatly and accurately. Corners shall be square and true. Flame cutting of ferrous metals may be employed using tips suitable for the thickness instead of shearing or sawing. Burned surfaces of flame-cut material shall be ground or machined sufficiently to remove ash and cooling checks. Aluminum alloys up to 0.500-inch in thickness may be cut by shearing; above 0.500 inch in thickness, alloys shall be cut by common or high-speed saws. All bends of a major character shall be made with metal dies or fixtures in order to insure uniformity of size and shape. Precautions shall be taken to avoid overheating, and heated metal shall be allowed to cool slowly.

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3.14.2 Castings. All castings shall be free of blowholes, slag inclusions, and other imperfections which could impair the serviceability or life of the parts.

3.14.3 Bolt and rivet holes. Holes in aluminum alloys shall be made by any one of the following methods:

a. Holes may be subpunched or subdrilled undersize and reamed to finish size upon assembly.

b. Holes may be drilled to size at assembly.

c. Holes may be drilled to size prior to assembly with accurately mated steel drilling jigs. The jigs shall be equipped with hardened steel bushings to insure the correct alinement and location of the holes. All burrs and rough edges shall be removed from the holes prior to final assembly.

3.14.4 Bolted connections. Boltholes shall be accurately punched or drilled. Washers or lockwashers shall be provided where necessary and all bolts, nuts, and screws shall be tight. Self-tapping sheet metal screws shall not be used. Drive screws may be used on the identification and instruction plates.

3.14.5 Riveted connections. Rivet holes shall be accurately punched or drilled. Rivets shall completely fill the holes. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

3.14.6 Welders and welding.

3.14.6.1 Welders. Before assigning any welder or welding operator to welding work covered by this specification, the contractor shall be responsible for certifying that the welder or welding operator has passed qualification tests as prescribed by either of the following listed codes for the type of welding operations to be performed and that such qualification is effective as defined by the particular code:

ANSI/ASME - Boiler and Pressure Vessel Code, Section IX, Welding and Brazing Qualifications.
AWS B2.1 - Standard for Welding Procedure and Performance Qualification.

Certification shall be made available for review by the contracting officer or designated representitive. Contractors who make only horizontal or flat welds need not qualify welders for "all position welding".

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3.14.6.2 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease, mill scale that can be removed by chipping and wire brushing, and other foreign matter. Parent materials, weld filler metals and fabrication techniques shall be as required to enable the unit to conform to the examination and test requirements specified in section 4. Parts to be joined by fillet welds shall be brought into as close contact as possible and in no event shall be separated by more than 0.1875-inch unless appropriate bridging techniques are used. Welds shall be free from discontinuities such as base metal cracks adjacent to or behind welds, cracks in weld metal, undercutting of base metal, lack of fusion with parent metal or between parts, lack of penetration, lack of prescribed fit, or splatter. Unless otherwise specified (see 6.2), the welding process used in fabrication of the unit shall be at the option of the contractor.

3.14.6.3 Aluminum welding. The aluminum surfaces of parts to be welded shall be brushed with a stainless steel brush until free from scale, paint, grease, oxide film, and other foreign matter. The welding shall be done by the inert gas, shielded arc method. Welds may be ground, filed, buffed, or chipped but shall not be hammered.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

c. Inspection of packaging (see 4.6).

4.3 First article inspection.

4.3.1 First article examination. The first article shall be examined as specified in 4.5.1. The presence of one or more defects shall be cause for rejection.

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4.3.2 First article tests. The first article shall be tested as specified in 4.5.2, in the order shown. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105.

4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined as specified in 4.5.1. AQL shall be 1.0 percent defective for major defects and 4.0 percent defective for minor defects.

4.4.3 Tests. Samples selected in accordance with 4.4.1 shall be tested as specified in 4.5.2. AQL shall be 1.0 percent defective.

4.5 Inspection procedure.

4.5.1 Examination. The unit shall be examined as specified herein for the following defects:

Examination schedule.

Number	Defects	Requirement Paragraph
Major		
101.	Material not as specified.	3.3
102.	Materials not resistant to corrosion and deterioration or treated to be made resistant to corrosion and deterioration.	3.3.1
103.	Dissimilar metals not in accordance with MIL-STD-889.	3.3.1.1
104.	Material and material finish or treatment information not available.	3.3.1.2
105.	Used, rebuilt or remanufactured components, pieces or parts incorporated in unit.	3.3.2
106.	Corrosion-resistant material used does not conform to AISI type corrosion-resistant steel.	3.3.3
107.	The design of the unit is not as specified.	3.4
108.	Fasteners loose, not as specified.	3.4.1
109.	Means for lubricating all surfaces requiring lubrication, not provided as specified.	3.4.2
110.	Lubrication fittings not as specified, and are not located in a protected position nor accessible to a grease gun.	3.5.2.1

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111.	Lubricant pressure release devices not provided as specified to prevent damage to grease seals or other parts	3.5.2.1.1
112.	Lubricant reservoir not equipped with a dipstick or sight gage as specified.	3.5.2.2
113.	Belt tension not adjustable or belt inaccessible for adjustment.	3.6
114.	Safety disk is not rated for the pressure from the cylinder head at specified range.	3.8.1
115.	Safety guard provided does not deflect flying particles.	3.8.1
116.	The bypass system does not protect the charging hose from blockage.	3.8.2
117.	Packing nut or plate not adjustable.	3.10.2.1
118.	Metal fabrication is not free from kinks, and sharp bends. Corners are not square and true. Shearing and chipping is not neat and accurate. Burned surfaces of flame-cut material is not ground or machined to remove ash and cooling checks.	3.14.1
119.	Casting is not free of blowholes, and slag inclusions.	3.14.2
120.	All burrs and rough edges have not been removed from bolt and riveted holes prior to final assembly as specified.	3.14.3
121.	Bolts used on bolted connections are missing washers or lockwasher and screws used are not tight.	3.14.4
122.	Rivets do not completely fill the hole of riveted connection. Rivets are not in full contact with the surface member.	3.14.5
123.	The contractor does not have certification that welders and welding operators have passed qualification tests as specified.	3.14.6.1
124.	The surfaces of parts welded are not free from rust, scale, paint, grease, mill scale and other foreign matter. Welding exhibits cracks in weld metal, under cutting of base metal, lack of fusion with parent metal, and lack of prescribed fit; or welds that have been hammered.	3.14.6.2
125.	Aluminum welds were welded by process other than in inert gas, shielded arc method.	3.14.6.3
126.	Human factors not as specified.	3.7
127.	Fungus and moisture resistance not as specified.	3.9.1

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Minor

201.	Instruction plates or identification marking, missing or incorrect.	3.5.2.3, 3.13
202.	Components normally painted have not been treated and painted in accordance with MIL-T-704, type A.	3.12

4.5.2 Tests.

4.5.2.1 Conditions. Prior to test, the unit shall be lubricated with greases specified in 3.10.7.1. Greases shall be those designated for use in the ambient temperatures at which the test is being performed.

4.5.2.2 Performance. Connect the unit to the charging and service cylinders with the hoses and service manifolding specified in 3.10.3 and charge not less than five service cylinders. During the cylinder filling operation, determine the operation and positioning of the tilt rack. Inability of the tilt rack to hold, invert, latch cylinders in place and return to the starting position, inability of the unit to transfer carbon dioxide at the rate specified in 3.10.2 or a malfunction or leakage that cannot be eliminated by making immediate integral adjustments shall constitute failure of this test.

4.5.2.3 Reliability/maintenance. Operate the pump for a total of 175 hours against a minimum back pressure of 350 psig above the inlet pressure. Perform scheduled maintenance after the first 100 hours of operation as prescribed by suggested maintenance schedule. Recirculation of the carbon dioxide through the pumping circuit is permitted during this test. Performance of the pump in transferring carbon dioxide shall be checked during the first 3 hours and at 24-hour intervals thereafter. The pump shall be stopped and restarted prior to each transfer operation. Storage cylinder shall be 50 lb capacity and service cylinder shall be 15 lb capacity. Inability of the pump to transfer carbon dioxide at the pressures and rates specified in 3.10.2 shall be considered a failure if more than simple adjustments are required to correct the shortcoming. Occurrence of one or more such failures will constitute failure of this test.

4.5.2.4 Bypass. Close the charging hose valve and connect the unit to a full, 50 lb, carbon dioxide cylinder. Open the cylinder valve and allow the carbon dioxide to pressurize the entire system. Start the pump and measure the pump outlet pressure. Any pressure in excess of 1,100 psig above the inlet pressure shall constitute failure of this test.

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4.5.2.5 High pressure. Subject the complete unit, including the hoses and fitting to a pressure of 3,000 psig. Damage to the unit, failure of any component or leakage of the system shall constitute failure of this test.

4.5.2.6 Noise level test. Noise levels shall be measured in accordance with MIL-STD-1474 requirements and reported in the format indicated by MIL-STD-1474, Figure 7. As a minimum: noise levels shall be measured when equipment is operating under full load, MIL-STD-1474, paragraph 5.1.2.1.4 contours shall be taken at not fewer than 12 equal (horizontal) arc increments, one increment shall include data from the noisiest position, additionally, the noise level at the typical operating position shall be provided as dB(A) level. Failure to comply with MIL-STD-1474 provisions shall constitute failure of this test.

4.5.2.7 Electromagnetic interference. Unless otherwise specified (see 6.2), the first article shall be tested to determine conformance with 3.11. The contractor shall furnish the contracting officer a report of tests required by MIL-STD-461. Disapproval of the report shall constitute failure of this test.

4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects. AQL shall be 1.0 percent defective.

126.	Preservation applied to surfaces of components that will contact carbon dioxide for level A or commercial.
127.	Pump not preserved and crankcase not drained as specified for level A.
128.	Openings into the pump not sealed as specified for level A.
129.	Drivebelt not removed or released from tension for level A.
130.	Unpainted pulleys and gears not coated with primer as specified for level A.
131.	Electric motor not preserved in accordance with the referenced document for level A.
132.	Hose not coiled and fittings not preserved in accordance with the referenced document for level A.
133.	Tilt rack, when assembled with threaded fasteners, not disassembled to attain minimum cube as specified for level A.

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134. Removed fasteners not replaced in mating parts or preserved as specified for level A.
135. Maintenance tools not preserved in accordance with the referenced document for level A.
136. Consolidation not as specified for level A.
137. Preservation and packing not in accordance with the referenced document specified for commercial.
138. Box not in accordance with the referenced document specified for level A or B.
139. Components not positioned within the box to provide the minimum cube for level A or B.
140. Blocking, bracing, and anchoring not in accordance with the referenced document for level A or B.
141. Marking missing, illegible, incorrect or incomplete for level A, B, or commercial.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or commercial as specified (see 6.2).

5.1.1 Caution. Surfaces of components that will contact carbon dioxide shall not be coated with any preservatives.

5.1.2 Level A.

5.1.2.1 Preservatives. Preservatives specified herein shall conform to the applicable specifications listed in and shall be applied in accordance with MIL-P-116.

5.1.2.2 Unprotected surfaces. Unpainted exterior metal surfaces of components requiring the application of contact preservative in accordance with MIL-P-116 and not specifically specified herein, shall be coated with type P-1 preservative.

5.1.2.3 Pump. The pump crankcase shall be filled to the operating level with oil conforming to MIL-L-21260, grade 30. The pump shall be actuated to assure coating of all interior parts and surfaces with the preservative and then drained. Openings into the pump shall be sealed with plastic or metal plugs.

5.1.2.4 Drive belt. The drive belt shall be removed or released from tension. If the drive belt is removed, it shall be preserved in accordance with MIL-P-116, method IC-1 or IC-3.

5.1.2.5 Pulleys and gears. Unpainted pulleys and gears shall be coated with primer conforming to TT-P-664.

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5.1.2.6 Electric motor. The electric motor shall be preserved in accordance with the level A requirements of MIL-E-16298, using the alternate method as specified for assembled machines. The attached electrical cable shall be coiled around the unit and secured with cotton tape or twine.

5.1.2.7 Hose and fittings. Each hose shall be coiled and the coil secured with cotton tape. The hoses and fittings shall be individually preserved in accordance with MIL-P-116, method IC-1 or IC-3.

5.1.2.8 Tilt rack. The tilt rack, when assembled with threaded fasteners, shall be disassembled as practicable to attain the minimum cube as required in 5.2.1. Removed fasteners shall be replaced in one of the mating parts or preserved together in accordance with MIL-P-116, method IC-1 or IC-3.

5.1.2.9 Repair parts. The preservative application criteria and applicable method(s) of preservation of MIL-P-116 shall be used to preserve repair parts.

5.1.2.10 Maintenance tools. Maintenance tools shall be preserved in accordance with the level A requirements of PPP-P-40.

5.1.2.11 Consolidation. The drive belt when removed, hoses and fittings, removed fasteners, repair parts and maintenance tools shall be consolidated together in the toolbox when space permits. Items not fitting in the toolbox or when a toolbox is not specified shall be consolidated together in a close-fitting box conforming to PPP-B-636, W5c. The box shall be closed and sealed in accordance with Method V of the appendix to the box specification.

5.1.3 Commercial. Each complete recharging unit shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B, or commercial as specified (see 6.2).

5.2.1 Level A. Each complete recharging unit shall be packed in a close-fitting box conforming to PPP-B-601, overseas type, style optional. The components shall be positioned in the box in a manner providing minimum cubage and shall be blocked, braced, and anchored in a manner to prevent free movement and damage in accordance with MIL-STD-1186. Box closure and strapping shall be in accordance with the appendix to the box specification.

5.2.2 Level B. Each complete recharging unit shall be packed as specified in 5.2.1 except the box shall be domestic type.

5.2.3 Commercial. Each complete recharging unit shall be packed in accordance with ASTM D 3951.

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5.2.3 Commercial. Each complete recharging unit shall be packed in accordance with ASTM D 3951.

5.3 Marking.

5.3.1 Military. Marking for military levels of protection (level A or B) shall be in accordance with MIL-STD-129.

5.3.2 Commercial. Marking for commercial packaging shall be in accordance with ASTM D 3951 and in addition shall indicate the weight and cube on the shipping container.

6. NOTES

6.1 Intended use. The recharging unit is intended to be used for transferring carbon dioxide from storage cylinders to service cylinders.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. When a first article is required for inspection and approval, and the number of units required (see 3.2).

c. When a tilt rack and toolbox is not required (see 3.4, 3.10.5 and 3.10.6).

d. When instruction plates and identification marking shall be commercial (see 3.5.2.3 and 3.13).

e. Composition II should be required when the application of varnish conforming to MIL-V-173 is accomplished as specified in 3.9.1 within a geographical area required to comply with Rule 442 South Coast Air Quality Management District, El Monte, California.

f. When the motor is to be other than in conformance to CC-M-1807 (see 3.10.4.1).

g. When the motor is to be 220 volt or 440 volt ac, or 115 volt or 230 volt dc (see 3.10.4.1).

h. When a power supply cable is not required (see 3.10.4.3).

i. When a power supply cable other than as specified is required (see 3.10.4.3) for Navy procurements.

j. When the starter is to be MIL-C-2212 (see 3.10.4.4) for Navy procurements.

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k. When the electromagnetic interference is to be other than as specified (see 3.11).

l. The color of paint for the recharging unit (see 3.12).

m. When the welding process will not be at the option of the contractor (see 3.14.6.2).

n. When certification of compliance is acceptable in lieu of test (see 4.5.2.6 and 6.5).

o. Degree of preservation and packing required (see 5.1 and 5.2).

6.3 First article. When a first article inspection is required, the item should be a preproduction model. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, tests and approval of the first article test results and disposition of the documents first articles.

6.4 Lubricants. The contracting officer should furnish a list of military lubricants applicable to the recharging unit covered by this specification as contained in the Department of Defense Federal Supply Catalog Identification List C9100-IL, for FSC Group 91.

6.5 Electromagnetic interference. In lieu of testing as specified in 4.5.2.6, a certificate of compliance may be accepted from the contractor.

6.6 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, and supplier's maintenance and operation manual to be furnished with each unit.

6.7 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including any special tools), and instructions on shipment of units. A suggested paragraph is as follows:

"Shipment of units shall include repair parts, maintenance tools, operational instructions, and accessories, unless exceptions are provided elsewhere in the contract."

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Army - ME
Air Force - 99

Review activity:
DLA - GS
Air Force - 82

User activities:
Navy - MC

Preparing activity:
Army - ME

Project 3655-0048

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-R-12551D	Recharging Unit, Carbon Dioxide, Pumping Method

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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